Note

Sometimes imagined class intervals do not exist. For example, in case of marks obtained by the students in a test, we cannot go below zero and beyond maximum marks on the two sides. In such cases, the extreme line segments meet at the mid points of the vertical left and right sides of first and last rectangles respectively.

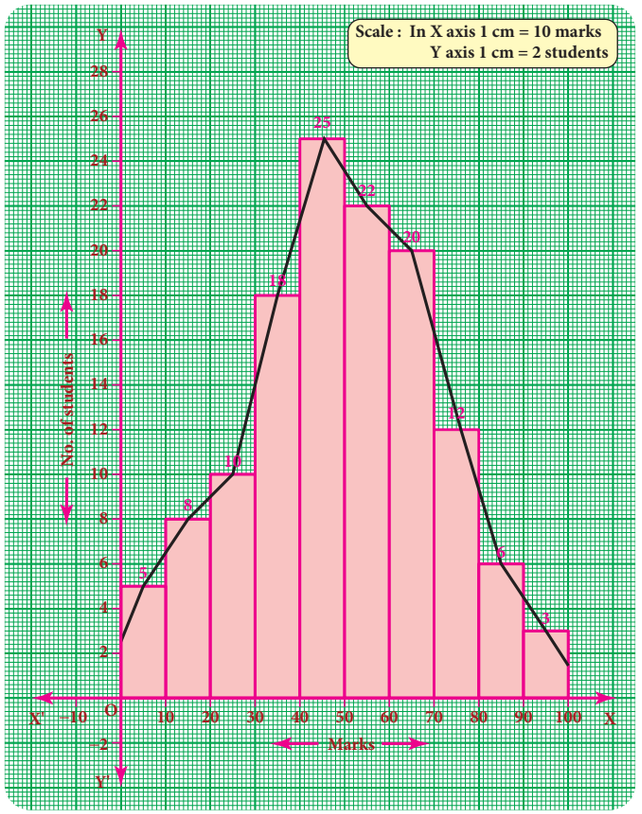
Example 6.10

Draw a frequency polygon for the following data using histogram.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Number of students	5	8	10	18	25	22	20	13	6	3

Solution:

Mark the class intervals along the x-axis and the number of students along the y-axis . Draw a histogram for the given data and mark the midpoints of the rectangles and join them by lines. We get frequency polygon. Note that the first and last edges of the frequency polygon meet at the mid points of the left and right vertical edges of first and last rectangles. Because imagined class intervals do not exist in the marks (refer the above note).



6.4.2 (ii) To draw a frequency polygon without using a histogram:

(1) Find the midpoints of the class intervals and tabulate it.

(2) Mark the midpoints of the class intervals on x-axis and frequencies on y-axis.

(3) Plot the points corresponding to the frequencies at each midpoints.

(4) Join the points using a ruler, to get the frequency polygon.

Example 6.11

Draw a frequency polygon for the following data without using histogram.

Class interval (Marks)	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Frequency	4	6	8	12	10	14	5	7

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